phaeno

WHITE PAPER

PSEQ WHITE PAPER SERIES · PAPER 3 OF 4

PSeq Data Pipeline

From raw FASTQ files to traceable whole-molecule RNA records

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Executive summary

The PSeq data pipeline is designed to transform paired-end short reads from a tagged RNA library into molecule-specific transcript records. It begins by locating the PSeq marker tag, recovering the Source Molecule Identifier (SMID) and strand signal, and grouping reads that originated from one tagged reverse transcript. Within each molecular bin, the pipeline identifies and retrieves the sequence of the source gene, assembles sequence, creates alignment products, and records the evidence chain in a run-specific SQL database.

This paper describes the computational architecture, data products, deployment model, and provenance strategy. It does not present the validation results themselves; the initial library and pipeline evidence is consolidated in Paper 4 of this series.

Computational objective: Turn a population of tagged short reads into independently inspectable molecular records without losing the connection to raw sequence and quality evidence.

Input model

The pipeline consumes raw paired-end FASTQ files directly from the sequencer together with run and sample metadata. Read 1 is expected to begin with a recognizable PSeq marker tag containing wrappers, a SMID, invariant check bases, and strand-sense information. The remainder of Read 1 and all or most of Read 2 contain cDNA fragment sequence.

The SMID changes the unit of computation. Rather than attempting to reconstruct transcript populations from all reads at once, the system first partitions read pairs into molecule-specific bins. Gene identification and retrieval, assembly, alignment, and generation of the consensus sequence can then operate within each bin.

Pipeline stages

1. Upload and register raw data. Receive the FASTQ files and associate them with client, sample, instrument, and PSeq run metadata.
2. Localize marker blocks. Find the wrapper and check-base landmarks expected at the start of Read 1.
3. Recover molecular identity. Parse the SMID and strand discriminator from valid marker tags.
4. Cluster read pairs by SMID. Place reads carrying the same source-molecule code into a molecule-specific file or logical bin.
5. Trim tag sequence. Remove marker and adaptor sequence so that downstream searches operate on cDNA sequence.
6. Deduplicate reads. Identify redundant trimmed records while retaining the evidence needed to audit the transformation.
7. Identify the source gene. Search trimmed reads and derived k-mers against an appropriate local transcript reference and retrieve the leading source-gene sequence.
8. Assemble the molecule. Perform de novo contig assembly and reference-guided alignment within the SMID bin.
9. Generate analysis products. Create FASTA, BAM, consensus, transcript-match, and summary records.
10. Populate the run database. Persist run, molecule, intermediate, and final records in the sample-run SQL database.

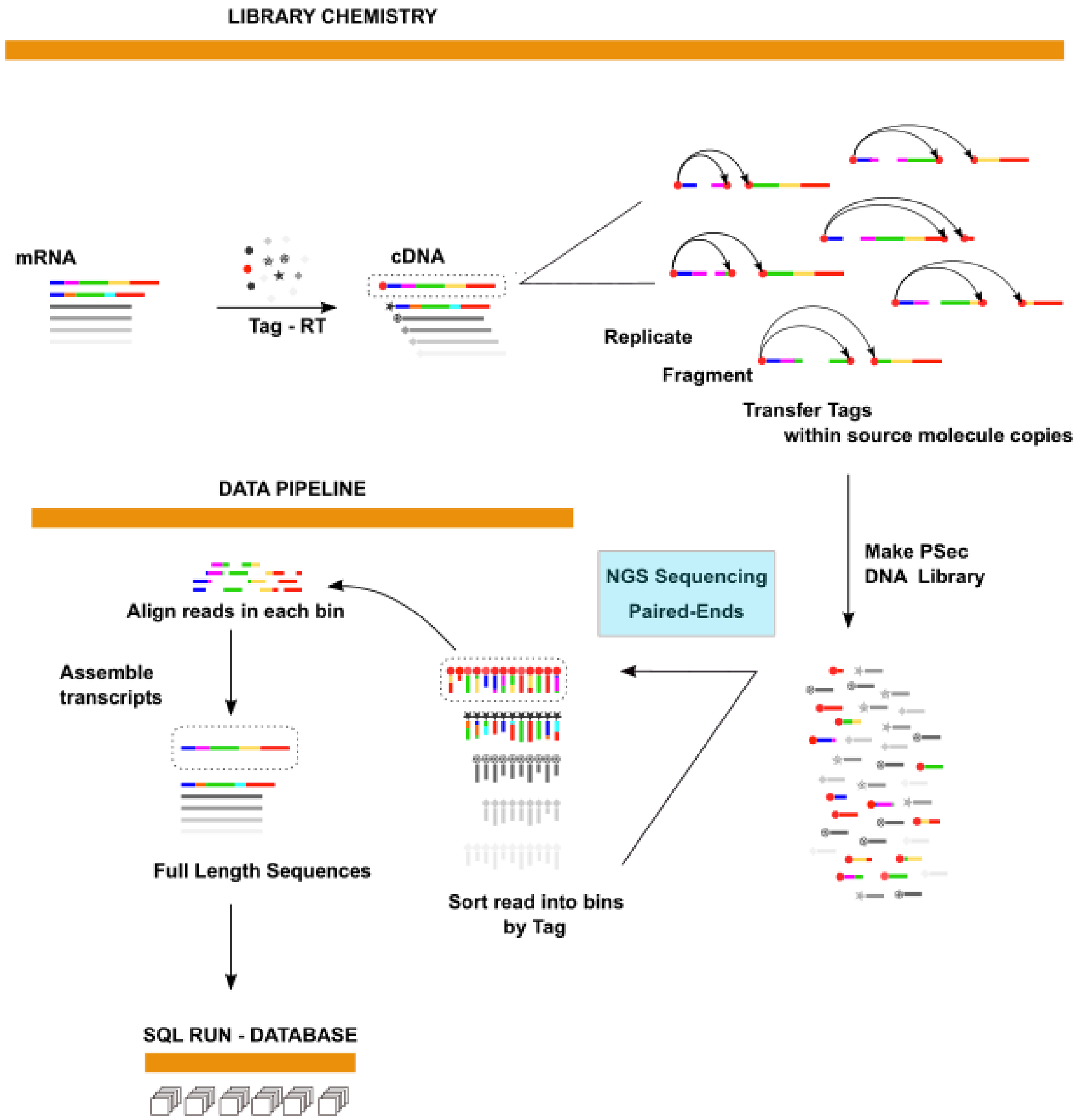

Figure 1. The data-pipeline portion of the integrated PSeq workflow groups reads by tag, assembles transcript sequences, and records outputs in a run-specific SQL database.

Gene identification and molecule assembly

Reference Search

For each SMID bin, the described implementation searches trimmed reads and read-derived k-mers against a local reference database. The leading transcript or gene match provides a candidate source-gene identity and retrieves the gene sequence for downstream comparison. Appropriate reference database may include ENS curated human transcript ensembles, databases for commercial synthetic spike-in standards including as SIRV or ERCC panels, or specialty databases for alternative species or infectious agents such as RNA viruses.

De novo assembly

Preliminary assemblies are alignment maps of trimmed reads to reference transcripts (e.g., ENS transcript database), for gene identification and retrieval of source gene sequence. For this purpose, modifications of open-source Inchworm assembler (uninterrupted contigs) and open-source Trinity assembler (single consensus sequence) are fast and slow options, respectively. De novo transcript assembly is performed by trimmed read alignments against the identified source gene, to capture included and skipped exons, and discover alternative translation initiation sequence and alternative polyadenylation sites that may lie within or outside of the Ref Seq annotated gene boundary.

Paper 4 of this series confirms that assembly implements true shotgun sequencing of individual transcripts with exceptionally high consensus sequencing accuracy. Intermediate steps in data assembly are recorded in the SMID file of the run-sample SQL database created by the pipeline.

Alignment and consensus

Trimmed reads within an SMID bin are aligned to the selected source-gene sequence and recorded in a BAM file. Alignment depth, exon coverage, splice-junction coverage and linkage continuity, mismatches, and base-quality evidence contribute to a molecule-specific consensus. Preliminary reference-transcript matches report length, percent coverage, and sequence agreement for comparison with annotated isoforms. Transcript alignment maps can be retrieved to profile population wide coverage, for example, as a function of transcript length.

Expected sequencing-read contract

The SQL database is the primary run deliverable described for PSeq v1. It is intended to preserve the result and the path used to produce it. The exact production schema remains an implementation specification, but the required information falls into several stable record layers

Record layer	Representative contents	Purpose
Client and sample	Client identifiers, sample description, project context	Connect molecular output to the submitted material
Run	Instrument/standardized run metadata, pipeline version, configuration	Connect run specifications to run reproducibility and optimization.
Raw evidence	FASTQ file references, read identifiers, quality scores	Maintain traceability to sequencer output
Molecule identity	SMID, strand sense, bin membership	Define the source-molecule unit of analysis
Gene assignment	Top transcript matches, gene ID, retrieved reference sequence	Document source gene selection
Assembly	Inchworm/Trinity contigs, reference transcript alignments	Preserve sequence construction intermediates.
Alignment	BAM output, coverage, splice-junction and mismatch evidence	Enable independent visual and quantitative review, and automated full-sample QC validation.
Final molecular record	FASTA, consensus sequence, summary fields, external gene links	Support client reports and downstream analysis

Provenance as a first-class requirement

A molecular consensus is more useful when its supporting evidence remains easily accessible. The PSeq data model is therefore intended to preserve the relationships among a consensus base, the aligned fragments that cover that position, the underlying FASTQ records including original base-quality scores. The same principle applies at higher levels: a gene assignment should point to the searches that supported it, and each splice junction should point to the molecule-specific alignments that span it.

This provenance model supports two modes of review. Individual molecules can be inspected manually using open-source alignment viewers such as IGV, while larger populations can be queried programmatically for quality-control summaries, unusual events, and for the generation of client reports.

Deployment and scaling

The initial architecture is installed for cloud-based implementation on Amazon EC2 processors. Each environment contains the pipeline software, local reference databases, and the run-specific SQL database populated during pipeline operation. Work can be scaled by increasing the number of processing cores within an instance and or by distributing work across multiple instances.

The pipeline operation targets a 6 to 12-hour analysis cycle; however shorter run times are likely feasible.. That target should remain explicitly qualified until runtime distributions are reported across defined run sizes, reference databases, and compute optimization in optional pipeline versions.

> **Operational Requirement:** Treat the tag sequence specification, library protocol, marker parser, and SMID/strand data model as one integrated interface. A change in any one component should trigger compatibility review across the others.

Client reporting and downstream computation

The run-specific SQL database is intended to support client-facing reports without discarding molecule-level detail. A report can summarize gene inventories, isoforms, sequence variants, novel isoforms, and quality indicators while preserving links back to each supporting molecular record.

The structured record set may also support machine-learning and natural-language SQL query execution for specific applications. Those are forward-looking uses rather than validation claims. Before describing PSeq data as AI-ready, the schema, field semantics, completeness rules, error states, and population-level quality metrics will be formally quantified and tested.

Validation interface

The pipeline itself must be validated as an integration of library chemistry, conventional NGS sequencing and automated computational utilities. Software validation addresses deterministic processing, error handling, database integrity, and record completeness. Scientific validation addresses whether SMID bins correspond to authentic source molecules, whether gene assignments are correct, whether assemblies recover accurate sequences and isoforms. QC reviews should penetrate records for each molecule discovered in each run and ascertain the extent to which reported confidence tracks observed error.

Paper 4 in this series documents initial evidence supporting well behaved integration of structural information in DNA sequencing libraries, NGS paired-end sequencing and the data pipeline utilities that consume raw FASTQ data executing the sequencer. A case study of a single RNA transcript documents data assembly and the and the traceability of all steps from raw data files through assembly and final consensus sequence. Typical experimental runs-SQL databases will encompass hundreds of thousands to hundreds of millions of transcript specific records.

A broader validation program will require reference standards, replicates, truth sets, and pre-specified performance metrics, as set out in that document.

Conclusion

The PSeq pipeline is designed around a simple computational advantage created by PSeq libraries: fragments can be grouped by experimentally assigned source-molecule identity before transcript reconstruction. The pipeline then combines marker parsing, read clustering, gene identification and retrieval, assembly, alignment, consensus generation, and assembles a provenance-rich SQL data product. Its distinguishing feature is not any one algorithm, but the continuity of evidence from raw read to independently inspectable molecular record.

Notes

1. Reference-database examples in the source report include a curated ENS transcript collection and synthetic SIRV or ERCC spike-in standards; production references and versions must be controlled.
2. IGV is an external open-source alignment viewer: https://igv.org/.
3. Runtime, throughput, and data-volume statements are design targets from the internal PSeq v1 overview and require formal benchmarking.

PSeq white paper series

PAPER 1	PAPER 2	PAPER 3	PAPER 4
PSeq: Whole-Molecule Phased RNA Sequencing on Standard Short-Read NGS	PSeq Molecular Tagging and Library Architecture	**PSeq Data Pipeline**	Initial Technical Validation of the PSeq Whole-Molecule RNA Sequencing Platform.